BITP 1123

DATA STRUCTURE AND ALGORITHM

PROJECT: HOTEL BOOKING SYSTEM

GROUP MEMBER:

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Add Guest and load data from file:

```
Guest data loaded from file.
Welcome to Sunshine Hotel System

==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
------------------------------------------------------------
Enter your choice (1-4): 1
Enter your name: Alicia Wong
Enter your phone number: 017-2938475
Enter preferred room number: 205
Enter booking date (e.g., 2025-06-22): 2025-07-03
Guest booking confirmed!

==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
------------------------------------------------------------
Enter your choice (1-4): 1
Enter your name: Farhan Ismail
Enter your phone number: 011-8329471
Enter preferred room number: 102
Enter booking date (e.g., 2025-06-22): 2025-06-28
Guest booking confirmed!
```

```
==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
------------------------------------------------------------
Enter your choice (1-4): 1
Enter your name: Chloe Tan
Enter your phone number: 018-2234678
Enter preferred room number: 310
Enter booking date (e.g., 2025-06-22): 2025-06-25
Guest booking confirmed!

==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
------------------------------------------------------------
Enter your choice (1-4): 1
Enter your name: Kevin Low
Enter your phone number: 016-9842310
Enter preferred room number: 107
Enter booking date (e.g., 2025-06-22): 2025-07-01
Guest booking confirmed!
```

```
==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
------------------------------------------------------------
Enter your choice (1-4): 1
Enter your name: Nurul Afiqah
Enter your phone number: 019-8892374
Enter preferred room number: 303
Enter booking date (e.g., 2025-06-22): 2025-06-30
Guest booking confirmed!
```

Search guest by Name:

```
Guest booking confirmed!

==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
------------------------------------------------------------
Enter your choice (1-4): 2
Enter your name to check booking: Farhan Ismail

--- Guest Found ---
Name : Farhan Ismail
Phone: 011-8329471
Room : 102
Date : 2025-06-28
----------------------
```

```
==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
------------------------------------------------------------
Enter your choice (1-4): 2
Enter your name to check booking: Nurul Afiqah

--- Guest Found ---
Name : Nurul Afiqah
Phone: 019-8892374
Room : 303
Date : 2025-06-30
----------------------
```

Search guest by room number and booking date:

```
==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
-------------------------------------------------------------
Enter your choice (1-4): 3
Enter room number: 310
Enter booking date (e.g., 2025-06-22): 2025-06-25

--- Guest Found ---
Name : Chloe Tan
Phone: 018-2234678
Room : 310
Date : 2025-06-25
-------------------------
```

```
==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
-------------------------------------------------------------
Enter your choice (1-4): 3
Enter room number: 205
Enter booking date (e.g., 2025-06-22): 2025-07-03

--- Guest Found ---
Name : Alicia Wong
Phone: 017-2938475
Room : 205
Date : 2025-07-03
-------------------------
```

Enter admin mode:

```
==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
-------------------------------------------------------------
Enter your choice (1-4): 99
Enter admin password: admin123
```

Display Guest in Linked List:

```
===================== ADMIN MENU =====================
  1. Display All Guests
  2. Edit Guest Details
  3. Delete Booking (Room & Date)
  4. Sort Guests by Name
  5. Sort Guests by Room Number
  6. Clear All Guest Data
  7. Exit Admin Menu
=========================================================
Enter your choice: 1

========== DISPLAY GUESTS MENU ==========
1. Display All Guests (Linked List)
2. Display Guests (Pre-Order Traversal)
3. Display Guests (In-Order Traversal)
4. Display Guests (Post-Order Traversal)
5. Return to Main Menu
==========================================
Enter your choice: 1
```

```
========================================
Displaying All Guests:
========================================
Name : Alicia Wong
Phone: 017-2938475
Room : 205
Date : 2025-07-03
-------------------------
Name : Farhan Ismail
Phone: 011-8329471
Room : 102
Date : 2025-06-28
-------------------------
Name : Chloe Tan
Phone: 018-2234678
Room : 310
Date : 2025-06-25
-------------------------
Name : Kevin Low
Phone: 016-9842310
Room : 107
Date : 2025-07-01
-------------------------
Name : Nurul Afiqah
Phone: 019-8892374
Room : 303
Date : 2025-06-30
-------------------------
Total guests: 5
```

Display guest in pre-order traversal:

```
========== DISPLAY GUESTS MENU ==========
1. Display All Guests (Linked List)
2. Display Guests (Pre-Order Traversal)
3. Display Guests (In-Order Traversal)
4. Display Guests (Post-Order Traversal)
5. Return to Main Menu
==========================================
Enter your choice: 2

=====================================
Displaying Guests (Pre-Order):
=====================================
Guests in Pre-Order Traversal:
Name : Alicia Wong
Phone: 017-2938475
Room : 205
Date : 2025-07-03
------------------------
Name : Farhan Ismail
Phone: 011-8329471
Room : 102
Date : 2025-06-28
------------------------
Name : Chloe Tan
Phone: 018-2234678
Room : 310
Date : 2025-06-25
------------------------
Name : Kevin Low
Phone: 016-9842310
Room : 107
Date : 2025-07-01
------------------------
Name : Nurul Afiqah
Phone: 019-8892374
Room : 303
Date : 2025-06-30
------------------------
Total guests: 5
```

Display guest in in-order traversal:

```
========== DISPLAY GUESTS MENU ==========
1. Display All Guests (Linked List)
2. Display Guests (Pre-Order Traversal)
3. Display Guests (In-Order Traversal)
4. Display Guests (Post-Order Traversal)
5. Return to Main Menu
==========================================
Enter your choice: 3

=====================================
Displaying Guests (In-Order):
=====================================
Guests in Alphabetical Order:
Name : Alicia Wong
Phone: 017-2938475
Room : 205
Date : 2025-07-03
------------------------
Name : Chloe Tan
Phone: 018-2234678
Room : 310
Date : 2025-06-25
------------------------
Name : Farhan Ismail
Phone: 011-8329471
Room : 102
Date : 2025-06-28
------------------------
Name : Kevin Low
Phone: 016-9842310
Room : 107
Date : 2025-07-01
------------------------
Name : Nurul Afiqah
Phone: 019-8892374
Room : 303
Date : 2025-06-30
------------------------
Total guests: 5
```

Display guest in post-order traversal:

```
========== DISPLAY GUESTS MENU ==========
1. Display All Guests (Linked List)
2. Display Guests (Pre-Order Traversal)
3. Display Guests (In-Order Traversal)
4. Display Guests (Post-Order Traversal)
5. Return to Main Menu
==========================================
Enter your choice: 4

=====================================
Displaying Guests (Post-Order):
=====================================
Guests in Post-Order Traversal:
Name : Chloe Tan
Phone: 018-2234678
Room : 310
Date : 2025-06-25
-------------------------
Name : Nurul Afiqah
Phone: 019-8892374
Room : 303
Date : 2025-06-30
-------------------------
Name : Kevin Low
Phone: 016-9842310
Room : 107
Date : 2025-07-01
-------------------------
Name : Farhan Ismail
Phone: 011-8329471
Room : 102
Date : 2025-06-28
-------------------------
Name : Alicia Wong
Phone: 017-2938475
Room : 205
Date : 2025-07-03
-------------------------
Total guests: 5
```

Edit guest:

```
========== DISPLAY GUESTS MENU ==========
1. Display All Guests (Linked List)
2. Display Guests (Pre-Order Traversal)
3. Display Guests (In-Order Traversal)
4. Display Guests (Post-Order Traversal)
5. Return to Main Menu
=========================================
Enter your choice: 5
Returning to Main Menu...

==================== ADMIN MENU ====================
  1. Display All Guests
  2. Edit Guest Details
  3. Delete Booking (Room & Date)
  4. Sort Guests by Name
  5. Sort Guests by Room Number
  6. Clear All Guest Data
  7. Exit Admin Menu
=======================================================
Enter your choice: 2
Enter guest name to edit: Farhan Ismail

--- Guest Found ---
Name : Farhan Ismail
Phone: 011-8329471
Room : 102

What would you like to edit?
1. Name
2. Phone
3. Room
4. Cancel
Enter choice: 3
Enter new room number: 103
```

After edit:

```
--------------------------
Name : Farhan Ismail
Phone: 011-8329471
Room : 103
Date : 2025-06-28
--------------------------
```

Delete entries:

```
===================== ADMIN MENU ====================
  1. Display All Guests
  2. Edit Guest Details
  3. Delete Booking (Room & Date)
  4. Sort Guests by Name
  5. Sort Guests by Room Number
  6. Clear All Guest Data
  7. Exit Admin Menu
=======================================================
Enter your choice: 3
Enter room number to delete: 103
Enter date: 2025-06-28
Guest in Room 103 on 2025-06-28 deleted successfully.
```

After delete:

```
=====================================
Displaying All Guests:
=====================================
Name : Alicia Wong
Phone: 017-2938475
Room : 205
Date : 2025-07-03
------------------------
Name : Chloe Tan
Phone: 018-2234678
Room : 310
Date : 2025-06-25
------------------------
Name : Kevin Low
Phone: 016-9842310
Room : 107
Date : 2025-07-01
------------------------
Name : Nurul Afiqah
Phone: 019-8892374
Room : 303
Date : 2025-06-30
------------------------
Total guests: 5
```

Sort guest by name:

```
==================== ADMIN MENU ====================
  1. Display All Guests
  2. Edit Guest Details
  3. Delete Booking (Room & Date)
  4. Sort Guests by Name
  5. Sort Guests by Room Number
  6. Clear All Guest Data
  7. Exit Admin Menu
=====================================================
Enter your choice: 4
Name : Alicia Wong
Phone: 017-2938475
Room : 205
Date : 2025-07-03
------------------------
Name : Chloe Tan
Phone: 018-2234678
Room : 310
Date : 2025-06-25
------------------------
Name : Farhan Ismail
Phone: 011-8329471
Room : 103
Date : 2025-06-28
------------------------
Name : Kevin Low
Phone: 016-9842310
Room : 107
Date : 2025-07-01
------------------------
Name : Nurul Afiqah
Phone: 019-8892374
Room : 303
Date : 2025-06-30
------------------------
Total guests: 5
```

Sort guest by room number:

```
==================== ADMIN MENU ====================
  1. Display All Guests
  2. Edit Guest Details
  3. Delete Booking (Room & Date)
  4. Sort Guests by Name
  5. Sort Guests by Room Number
  6. Clear All Guest Data
  7. Exit Admin Menu
=========================================================
Enter your choice: 5
Name : Farhan Ismail
Phone: 011-8329471
Room : 103
Date : 2025-06-28
------------------------
Name : Kevin Low
Phone: 016-9842310
Room : 107
Date : 2025-07-01
------------------------
Name : Alicia Wong
Phone: 017-2938475
Room : 205
Date : 2025-07-03
------------------------
Name : Nurul Afiqah
Phone: 019-8892374
Room : 303
Date : 2025-06-30
------------------------
Name : Chloe Tan
Phone: 018-2234678
Room : 310
Date : 2025-06-25
------------------------
Total guests: 5
```

Clear data:

```
==================== ADMIN MENU ====================
  1. Display All Guests
  2. Edit Guest Details
  3. Delete Booking (Room & Date)
  4. Sort Guests by Name
  5. Sort Guests by Room Number
  6. Clear All Guest Data
  7. Exit Admin Menu
=========================================================
Enter your choice: 6
Are you sure you want to clear ALL data? (Y/N): Y
All guest data cleared.
```

After clear data:

```
========================================
Displaying All Guests:
========================================
No guest records found.
```

Save data to file:

```
==================== ADMIN MENU ====================
  1. Display All Guests
  2. Edit Guest Details
  3. Delete Booking (Room & Date)
  4. Sort Guests by Name
  5. Sort Guests by Room Number
  6. Clear All Guest Data
  7. Exit Admin Menu
=========================================================
Enter your choice: 7
Exiting admin menu.

==================== SUNSHINE HOTEL BOOKING SYSTEM ====================
Welcome! Please choose an option below:

  1. Book a Room
  2. Check My Booking (Name)
  3. Check Room Booking (Room & Date)
  4. Exit
---------------------------------------------------------------
Enter your choice (1-4): 4
Guest data saved.
Thank you for choosing Sunshine Hotel. Have a pleasant stay.
```